SUSMITHA CHOWDARY

SUSMITHA CHOWDARY | Data Engineer | Python · SQL · PySpark · Databricks · Snowflake · Azure · AWS | Susmithachowdary67@gmail.com | 9132073787 | 1601 Scottsdale Ln, Austin, Texas, 78754

Data Engineer

Summary

Candidate aligned to Software Engineer, Data & Analytics Platform priorities including data quality. Data Engineer with 5 years of experience in healthcare and banking data, with experience developing and supporting banking ETL at Citi and owning Azure, Databricks, and Snowflake pipelines at Optum. Builds incremental ingestion, automated data-quality checks, and dimensional models that analytics teams rely on. Experienced with HIPAA-regulated PHI at Optum and with payment, transaction, fraud-analytics, and lending data, regulatory reporting, and data controls at Citi.

Experience

Data Engineer | Optum | Austin, TX | May 2025 – Present

Own production ingestion pipelines in Azure Data Factory and Azure Databricks, integrating claims, eligibility, and provider feeds into ADLS Gen2 and Delta Lake bronze, silver, and gold layers; designed a reusable, config-driven PySpark ingestion framework that cut onboarding time for a new source feed from ~2 weeks to ~4 days.

Built automated data-quality checks for duplicate claim lines, missing identifiers, invalid service dates, and eligibility mismatches, quarantining failed records before publication, and applied HIPAA-aligned PHI controls (column-level masking, role-based access, PHI-safe logging) in Snowflake and Azure.

Improved scheduled pipeline success rate from about 97% to 99.5% with dependency checks, bounded retries, and idempotent backfills that recover failed runs without duplicating data, and lowered Databricks compute cost through cluster right-sizing.

Delivered the Snowflake layer: loaded validated Parquet via external stages and COPY INTO, applied MERGE logic for new and adjusted claims, and modeled claims, member, and provider dimensions used by multiple analytics teams.

Maintain CI/CD and automated tests in Azure DevOps, review pull requests, mentor junior engineers, and take part in the on-call rotation.

Reduced runtime of the critical claims-processing workflow from about 2 hours to 75 minutes using incremental Delta merges, partition pruning, and resolution of skewed Spark joins.

Data Engineer | Citi | Pune, India | January 2020 – May 2023

Progressed from SQL transformations, Python validation, and source-to-target testing to owning batch ETL jobs in AWS Glue and PySpark, integrating Oracle extracts and file feeds through Amazon S3 landing and curated layers and processing ~5 million payment and transaction records per business day.

Delivered curated transaction datasets to the fraud analytics team for rule and model scoring, adding validation checks on transaction classifications and timestamps so alerts ran on complete, consistent data.

Prepared and reconciled datasets for monthly regulatory and risk reporting with lineage documentation and control checks for audit review, and implemented SCD Type 2 history for

customer and account dimensions to support point-in-time lending and credit-risk analysis.

Shortened the overnight batch window from about 5 hours to 3 hours through watermark-based incremental extraction and join optimization, and supported production jobs in the on-call rotation using CloudWatch logs, restart checkpoints, and runbooks.

Automated source-to-target reconciliation controls over record counts, transaction totals, and account balances, producing exception files for investigation and cutting daily manual verification from about 90 to 25 minutes.

Skills

AWS Glue, AWS: Amazon S3, Azure Databricks, Azure: Azure Data Factory, data controls, Data Engineering: ETL/ELT, Data Platforms: Snowflake, data quality, Databricks, Domain & Compliance: Healthcare claims, fraud analytics data, Glue Data Catalog, HIPAA/PHI handling, Languages: Python, lending and credit-risk data, payment and transaction data, SQL, SQL transformations, ADLS Gen2, Amazon Athena, Amazon CloudWatch, Apache Spark, automated testing, Azure DevOps, Bash, CI/CD, controlled backfills, COPY INTO, Delta Lake (medallion architecture), DevOps & Operations: Git, dimensional modeling, incremental ingestion, masking policies, MERGE, on-call incident response, Oracle, pipeline monitoring, PySpark, reconciliation, regulatory reporting, role-based access, SCD Type 2, Snowflake: External stages

Education

Master of Science in Computer Science | University of Central Missouri | Warrensburg, MO | Jan 2024 – Dec 2025